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Oefeningenreeks 5G nr. 1.10

Bereken het volume V dat men bekomt door de functie $f(x) = \sqrt{1 + \sin^2 x}$ op het interval $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ om de X -as te wentelen.

$$\begin{aligned} V &= \pi \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(\sqrt{1 + \sin^2 x} \right)^2 dx \\ &= \pi \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (1 + \sin^2 x) dx \\ &= \pi \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(1 + \frac{1 - \cos(2x)}{2} \right) dx \\ &= \pi \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(\frac{3}{2} - \frac{\cos(2x)}{2} \right) dx \\ &= \pi \left[\frac{3}{2}x - \frac{\sin(2x)}{4} \right]_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \\ &= \pi \left(\left(\frac{3\pi}{4} - \frac{\sin(\pi)}{4} \right) - \left(-\frac{3\pi}{4} - \frac{\sin(-\pi)}{4} \right) \right) \\ &= \pi \left(\frac{3\pi}{4} - 0 - \left(-\frac{3\pi}{4} \right) + 0 \right) \\ &= \pi \left(\frac{6\pi}{4} \right) \\ &= \frac{3\pi^2}{2} \end{aligned}$$

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References

- [1] grafiek gemaakt in desmos